

```
import java.util.ArrayList;
import java.util.Scanner;
class GradeCalculator
{
    public static void main (String[] args)
    {
        Scanner scanner= new Scanner(System.in);
        ArrayList<Double>marks= new ArrayList<>();
        System.out.print("Enter the number of subjects");
        int totalSubjects=scanner.nextInt();
        for (int i=1;i<=totalSubjects;i++)
        {
            System.out.print("Enter marks for subject"+ i +"
                               (out of 100):");
            double mark=scanner.nextDouble();
            while(mark<0||mark>100)
            {
                System.out.print("Invalid marks! Please enter
                                   between 0 and 100");
                mark=scanner.nextDouble();
            }
            marks.add(mark);
        }
        double totalMarks=0;
        for(double m :marks)
        {
            totalMarks+=m; }
    }
}
```

```
double averagePercentage = totalMarks;
```

```
char grade;
```

```
if (averagePercentage>=85)
```

```
{
```

```
    grade ='A';
```

```
}
```

```
else if (averagePercentage>=70)
```

```
{
```

```
    grade ='B';
```

```
}
```

```
else if (averagePercentage>=50)
```

```
{
```

```
    grade = 'C';
```

```
}
```

```
else
```

```
{
```

```
    grade='F'; }
```

```
System.out.println("\n---Final Result---");
```

```
System.out.println("Total Marks Obtained."+  
totalMarks+"/"+"(totalSubjects*100));
```

```
System.out.printf("AveragePercentage:%2f%%\n",averagePercentage);
```

```
System.out.println("Assigned Grade:"+grade);
```

```
scanner.close();
```

```
}
```

```
}
```